

Class Activity - Generic methods and Class

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

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QUESTIONS

Q1 Swap Two Values
10 marks

Q2 Find the Maximum Mark
10 marks

Q3 Generic Class – Product
Inventory Holder
10 marks

3/3 answered

Q1 Swap Two Values 10 marks

Write a Java program that creates a generic method named swap() to interchange two values.
Create a generic method swap(T a, T b).
Call the method using:
Integer values
String values
Display the values before and after swapping.

Your Code

```
1 public class Main {  
2     public static <T> void swap(T a, T b) {  
3         System.out.println("Before swap: a=" + a + ", b=" + b);  
4         System.out.println("After swap : a=" + b + ", b=" + a);  
5     }  
6     public static void main(String[] args) {  
7         Integer i1 = 10, i2 = 80;  
8         swap(i1, i2);  
9         System.out.println();  
10        String s1 = "Hi", s2 = "swarna";  
11        swap(s1, s2);  
12    }  
13 }
```

Input
stdin input

Output
Before swap: a=10, b=80
After swap : a=80, b=10

Before swap: a=Hi, b=swarna
After swap : a=swarna, b=Hi

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3/3 answered

String product name
Double product price

Sample Output
Product ID: 101
Product Name: Wireless Mouse
Product Price: 799.50

Your Code

```
1 class Main {  
2     static class Inventory<T> {  
3         T item;  
4         Inventory(T item) {  
5             this.item = item;  
6         }  
7         void showItem(String label) {  
8             System.out.println(label + ": " + item);  
9         }  
10    }  
11    public static void main(String[] args) {  
12        Inventory<Integer> invId = new Inventory<>(101);  
13        Inventory<String> invName = new Inventory<>("Wireless Mouse");  
14        Inventory<Double> invPrice = new Inventory<>(799.50);  
15        invId.showItem("Product ID");  
16        invName.showItem("Product Name");  
17        invPrice.showItem("Product Price");  
18    }  
19 }
```

Input
stdin input

Output
Product ID: 101
Product Name: Wireless Mouse
Product Price: 799.5